4. System Architecture

GridPulse will use a modular layered architecture that connects simulated grid telemetry, data processing, AI prediction, safety validation, and the web dashboard.

System Flow:

Grid Simulator
↓
Telemetry Ingestion
↓
Analytical Pre-Processor
↓
AI Predictive Engine
↓
Safety Rule Enforcer
↓
Backend API & Database
↓
Interactive Web Dashboard

- **Grid Simulator:** Generates telemetry for the simulated 20-node grid, including voltage, load, generation, and node status.
- **Telemetry & Pre-Processor:** Receives, validates, and processes telemetry using 5-minute rolling averages and grid constraints.
- **AI Predictive Engine:** Forecasts demand spikes up to four hours ahead and identifies abnormal conditions.
- **Safety Rule Enforcer:** Validates AI recommendations against predefined node and cable capacity limits.
- **Backend & Database:** Manages APIs, processing services, and time-series telemetry storage.
- **Web Dashboard:** Displays grid topology, energy flow, forecasts, alerts, and simulation controls.

The architecture follows a **predict–validate–visualize** approach, keeping AI recommendations within predefined safety constraints.

5. Expected System Outcome

The completed GridPulse system will provide an interactive AI-assisted platform for monitoring and analyzing a simulated 20-node smart grid.

The system will support:

- Real-time grid and telemetry monitoring

- Interactive grid topology and energy-flow visualization
- Four-hour-ahead demand forecasting
- Abnormal condition and anomaly detection
- Grid capacity violation detection
- AI recommendation safety validation
- Grid-shock simulation
- Centralized dashboard-based monitoring

Overall, GridPulse will demonstrate how **AI-based prediction combined with deterministic safety rules** can support safer and more proactive smart-grid management. The system will remain a **simulation and decision-support platform** and will not directly control real-world electrical infrastructure.